

type indicating an 8-octet [IEEE 754-2019](#) double-precision value.

If the [nullable](#) qualifier is given, a TLV **Null** MAY be encoded in place of the number.

The allowed range of values can be constrained using the [range](#) qualifier. If omitted, the value is constrained by what the relevant TLV type can represent.

Float Example

```
set-value => FLOAT32 [ range 0..50 ]
```

B.3.4. SIGNED INTEGER / UNSIGNED INTEGER

Integer Syntax

```
SIGNED INTEGER [ qualifier ] { enum }  
UNSIGNED INTEGER [ qualifier ] { enum }
```

qualifier (optional):

[range](#)

[nullable](#)

enum:

identifier = int-value

SIGNED INTEGER declares an element that SHALL be encoded as a TLV [integer](#) with the [element type](#) indicating the integer is signed. Correspondingly, **UNSIGNED INTEGER** declares a TLV element that SHALL be encoded as a TLV [integer](#) with the [element type](#) indicating the integer is unsigned.

If the [nullable](#) qualifier is given, a TLV **Null** MAY be encoded in place of the integer.

The allowed range of values MAY be constrained using the [range](#) qualifier. If omitted, the value is constrained by what the relevant TLV type can represent.

SIGNED INTEGER and **UNSIGNED INTEGER** definitions MAY include a set of enumerated values ([enum](#)), each of which associates a textual name ([identifier](#)) with a constant integer value ([int-value](#)). Each value SHALL conform to the allowed range of values for the **SIGNED INTEGER** definition as given by its sign and any [range](#) qualifier. The presence of enumerated values SHALL NOT restrict senders to only encoding those values. Rather, enumerations merely give symbolic names to particular noteworthy values.

Integer Examples

```
sensor-value => SIGNED INTEGER [ range -100..100 ]
```

```
counter => UNSIGNED INTEGER [ range 32-bits ]
```